

# PASCAL DESOUZA

RESEARCH, DATA SCIENCE | DIGITAL TRANSFORMATION

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## SELECTED PROFESSIONAL EXPERIENCES

### **BSYSTEMS Limited, Accra, Ghana.**

**2020-2025**

- Research/Data Analyst
- Developed and automated full merchant reconciliation reports (RP & SR) using Python and Excel, covering deducted failed transactions, auto/manual reversals, and final reconciliation position across all months.
- Conducted data cleaning, matching, and risk analysis to track Direct Credits, Payouts, Send Money, Settlements, and balance snapshots while identifying over-reversals and merchant risk.

ANKH RESEARCH & DISCOVERY, Kumasi, Ghana.

- Web Administrator & Data Analyst
- Designed and developed responsive web interfaces and internal dashboards to support research data visualization and reporting workflows.
- Collaborated with cross-functional teams to integrate analytical tools and database connectivity for improved data accuracy and accessibility.
- Deployed and maintained web applications on secure hosting environments, ensuring system reliability and user-friendly performance.

CON-IMEDIA COMPANY LIMITED, Adenta, Ghana.

- Server Administrator | Business Operations
- Delivered strategic analysis and performance reports to optimize operational decisions that directly impacted client satisfaction and aligned stakeholder expectations.
- Coordinated process optimization initiatives, leveraging data insights to enhance workflow efficiency and user experience for improved service delivery.
- Prepared detailed reports on system performance and infrastructure, informing operational strategies that indirectly benefited end-user stability and experience.

SOFTMASTERS COMPANY LIMITED, Abelemkpe, Ghana.

- Server Administrator | Business Operations
- Coordinated with departments to build comprehensive proposals for products and services presented to prospective clients.
- Monitored team projects to ensure execution aligned with agreed timelines, deliverables, and quality standards
- Gathered and synthesized useful information from customer and competitor data to inform business strategy and designed and delivered presentations to prospective clients and internal executives to drive business growth
- Tracked and reported on the status of proposal components throughout the sales pipeline.
- Teaching Assistant, KNUST School of Business
- Course: Introduction to Programming
- Level: Undergraduate
- Program: BSc. Business Information Technology
- Teaching Assistant, KNUST School of Business
- Course: Project Management
- Level: Undergraduate
- Program: BSc. Business Administration

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## SELECTED DATA SCIENCE EXPERIENCES

### Amazon Product Reviews

2025

Data Scientist

- Processed and analysed Amazon customer review data to extract insights on product sentiment and rating distributions. Conducted text preprocessing and sentiment analysis in R, and developed classification models to distinguish positive from negative reviews, supporting product recommendation strategies.

### Credit Card Transactions Dataset

2025

Data Scientist

Performed fraud detection analysis on a large credit card transaction dataset with significant class imbalance. Applied data cleaning, SMOTE-based resampling, and machine learning models in R, including logistic regression and decision trees, to improve fraud identification precision and recall metrics. (<https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud>)

### EPA Environmental

2025

Data Scientist

- Cleaned and analysed a U.S. EPA air quality dataset sourced from Kaggle (<https://shorturl.at/epaAQI>) covering pollutant concentrations including NO<sub>2</sub>, SO<sub>2</sub>, CO, and Ozone across multiple U.S. states and counties. Performed exploratory data analysis and AQI trend analysis in R and Python, visualizing geographic and temporal patterns in air pollution to identify regions with persistently poor air quality. (<https://www.kaggle.com/datasets/saswatsethda/u-s-epa-air-quality-dataset>)

### Marketing Campaign

2024

Data Scientist

- Analysed a marketing campaign dataset to evaluate customer response rates, segmentation, and campaign effectiveness. Applied clustering techniques and logistic regression to identify high-value customer profiles and predict campaign conversion, supporting data-driven decisions for targeted marketing strategies.

### School Performance Dataset.

2024

Data Scientist

- Explored student academic performance data to identify factors influencing outcomes across subjects and demographics. Applied data cleaning, exploratory analysis, and predictive modelling in R to assess the effect of study habits, attendance, and parental engagement on student grades.

### Sleep and Mental Health Dataset.

2024

Data Scientist

- Analysed a dataset linking sleep quality and duration to mental health indicators including stress, anxiety, and depression scores. Cleaned and explored the data in python and R, applied correlation analysis and regression modelling to quantify the relationship between sleep patterns and mental wellness outcomes. (<https://www.cdc.gov/brfss/index.html>)

### Spotify Music Streaming Dataset

2024

Data Scientist

- Worked with a Spotify audio features dataset to explore patterns in song popularity, genre classification, and listener preferences. Applied feature selection, clustering, and classification models in R to predict track popularity and uncover the audio characteristics, tempo, danceability, energy, most associated with streaming success. (<https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset?select=dataset.csv>)

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**SELECTED AI-DRIVEN SOFTWARE ENGINEERING EXPERIENCES**

**Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana.** **2024-2025**

AI & Predictive Analytics Projects

Designed and implemented AI-driven predictive analytics systems using Python to forecast retail sales performance across multiple locations, integrating time-series and machine learning models (ARIMA, Random Forest, Gradient Boosting, Prophet) and engineering end-to-end pipelines including data preprocessing, feature engineering, model training, validation, and performance evaluation for operational optimisation and decision support

Developed an integrated intelligent forecasting framework combining sales, inventory, workforce allocation, promotions, weather, and external event datasets, applying unsupervised learning (K-Means clustering) for segmentation and building Power BI and Python dashboards to communicate predictive insights and support data-driven strategic planning

**UrbanGrocers Ltd. Predictive Analytics Simulation Project.** **2025-2025**

AI & Data Analytics Engineer (Industry Case Study)

Developed machine learning models to forecast product category demand using historical sales, inventory, workforce, and external economic event data across ten retail locations in Ghana. And also engineering predictive algorithms supporting inventory optimisation and workforce allocation decisions to minimise stock shortages and operational inefficiencies.

Implemented feature engineering strategies capturing seasonality, promotions, and environmental factors to improve forecasting accuracy and business intelligence outcomes.

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## SELECTED CERTIFICATIONS

- NCCRM Data Hub – Capacity Building Training (Certificate of Participation), 2026. Organised by GIZ – EU Support to ECOWAS on Peace, Security and Governance (EPSG) Project. Training covered digital tools and data management for peace, security and governance within the ECOWAS region.
- ASONG – Affordable Sustainability for Logistics Networks in Ghana (Certificate of Participation), 2025 Co-hosted by Kwame Nkrumah University of Science and Technology (KNUST) and Technische Universität Berlin. Training focused on sustainable logistics networks, digital technologies for sustainability, supply chain analytics, sustainability assessment and reporting, logistics network design, and data analysis & visualization using Power BI and Excel.
- CIP (Code in Place) – Certificate of Completion, 2025 Global online programming program organized by Stanford University, USA
- Completed multiple leadership and governance development programmes organized by the International Leadership Foundation (2018–2019), focusing on character formation, integrity, collaborative leadership, capacity building, coaching, communication, team-building, and conflict resolution, earning Certificates of Completion and Participation.

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## EDUCATION

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| <b>KWAME NKURUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY</b>              | <b>Kumasi, Ghana.</b> |
| <b>[MSc. Business and Data Analytics]</b>                               | <b>2024-2026</b>      |
| <b>KWAME NKURUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY</b>              | <b>Kumasi, Ghana.</b> |
| <b>[BSc. Business Administration • Business Information Technology]</b> | <b>2016-2020</b>      |

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## KEY SKILLS

- Data Analysis and Visualization: Excel, SQL, Power BI, Python, R
- Dashboard Development and KPI Reporting
- Business Operations and Process Optimization
- CRM Systems and Workflow Documentation
- Project Coordination and Stakeholder Engagement
- Remote Collaboration and Cross Functional Communication
- Workflow Automation and Continuous Improvement

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## TOOLS & PLATFORMS

Programming & Analytics: Python, R, SQL, Jupyter Notebook, JASP

Data Analysis & Visualization: Excel, Power BI, Data Visualization & Reporting

Web Technologies: HTML, CSS, JavaScript, WordPress

Hosting & Server Management: cPanel, WHM

Productivity & Collaboration: Microsoft Office Suite, Google Workspace, Microsoft Teams, Zoom

Design & Content Creation: Canva, Photoshop

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## REFERENCES

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